

Vector Databases & Embeddings for Developers

Understanding Vector Databases
and the Standard Retrieval Model



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Version Check



This course was created by using:

- C# 8
- Microsoft Azure free resources



Relevant Notes



A note on frameworks and libraries:

- The examples and demonstrations in this course can be replicated in virtually any programming language.



Demo



Achieving image retrieval from a vector database





Introduction to Retrieval Models



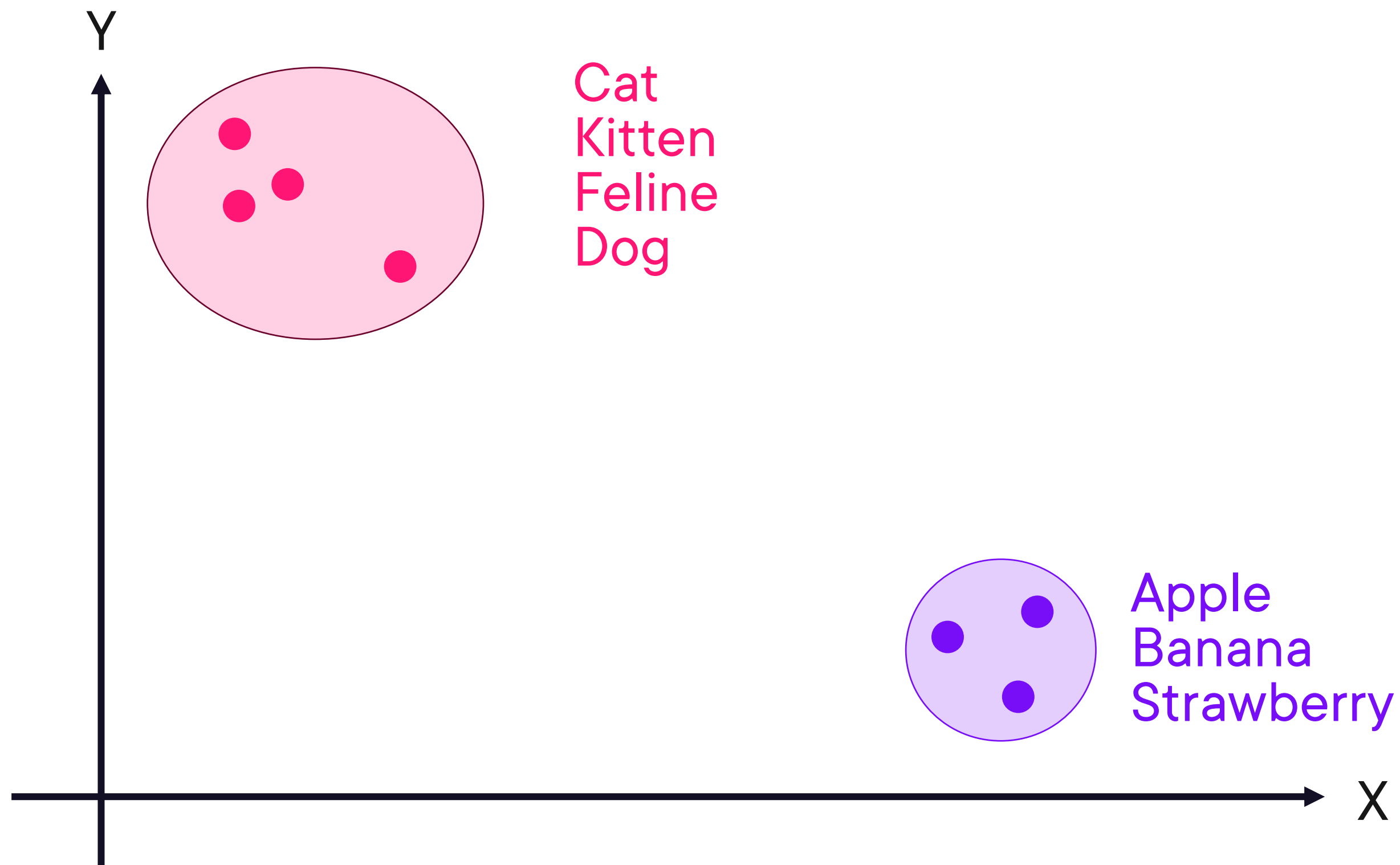
Classification of Elements in a 2 Dimensions Space



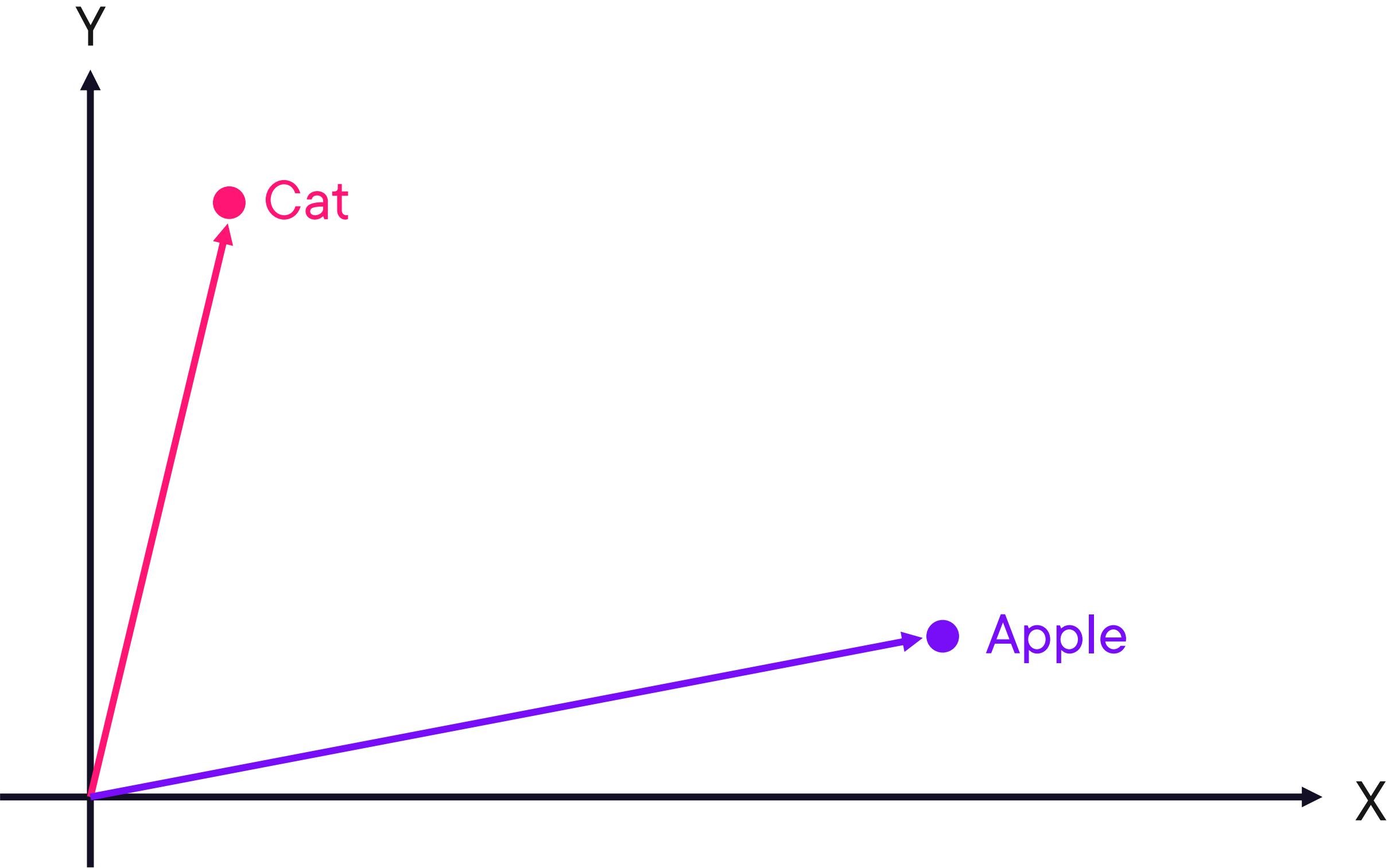
Classification of Elements in a 2 Dimensions Space



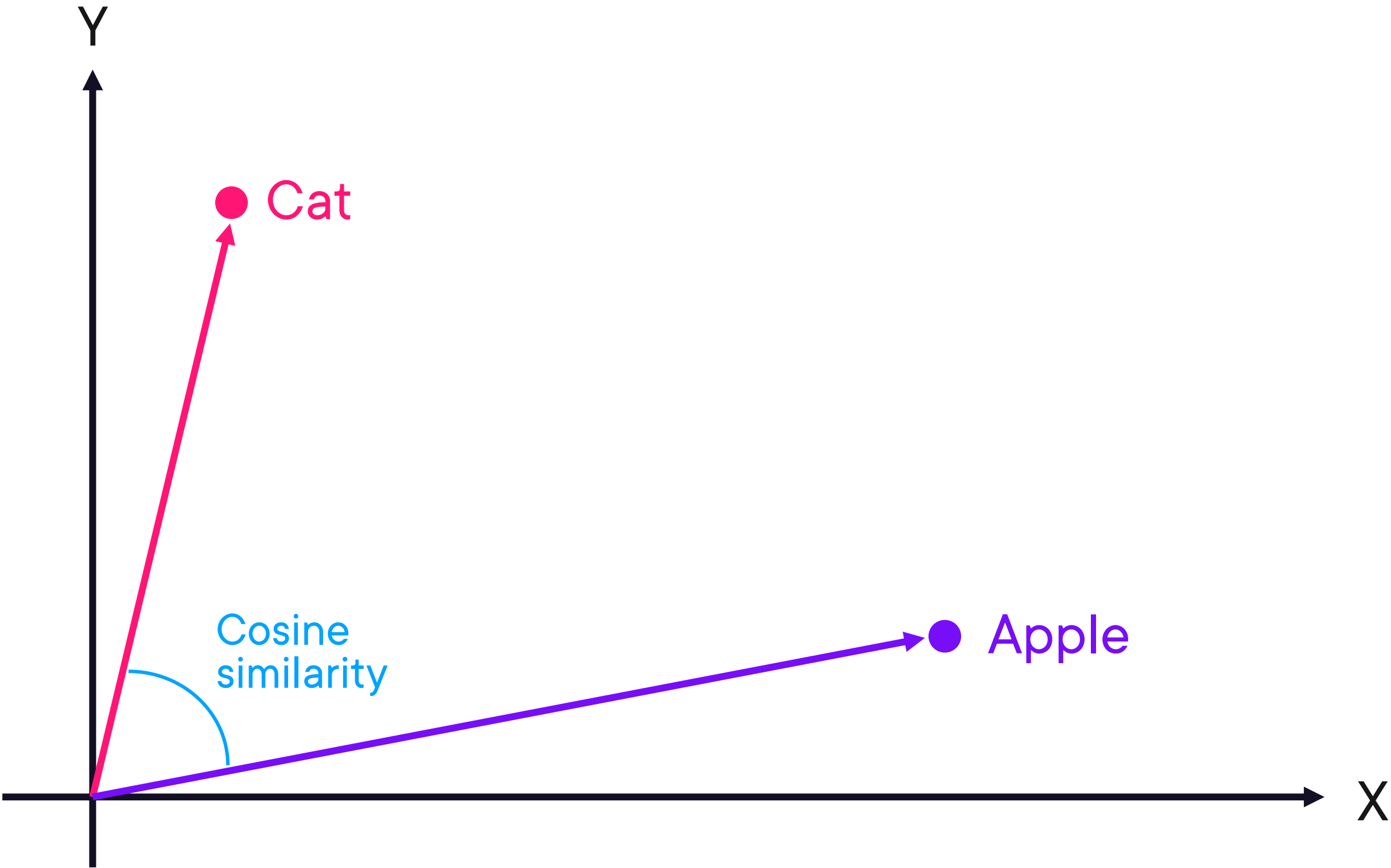
Classification of Elements in a 2 Dimensions Space



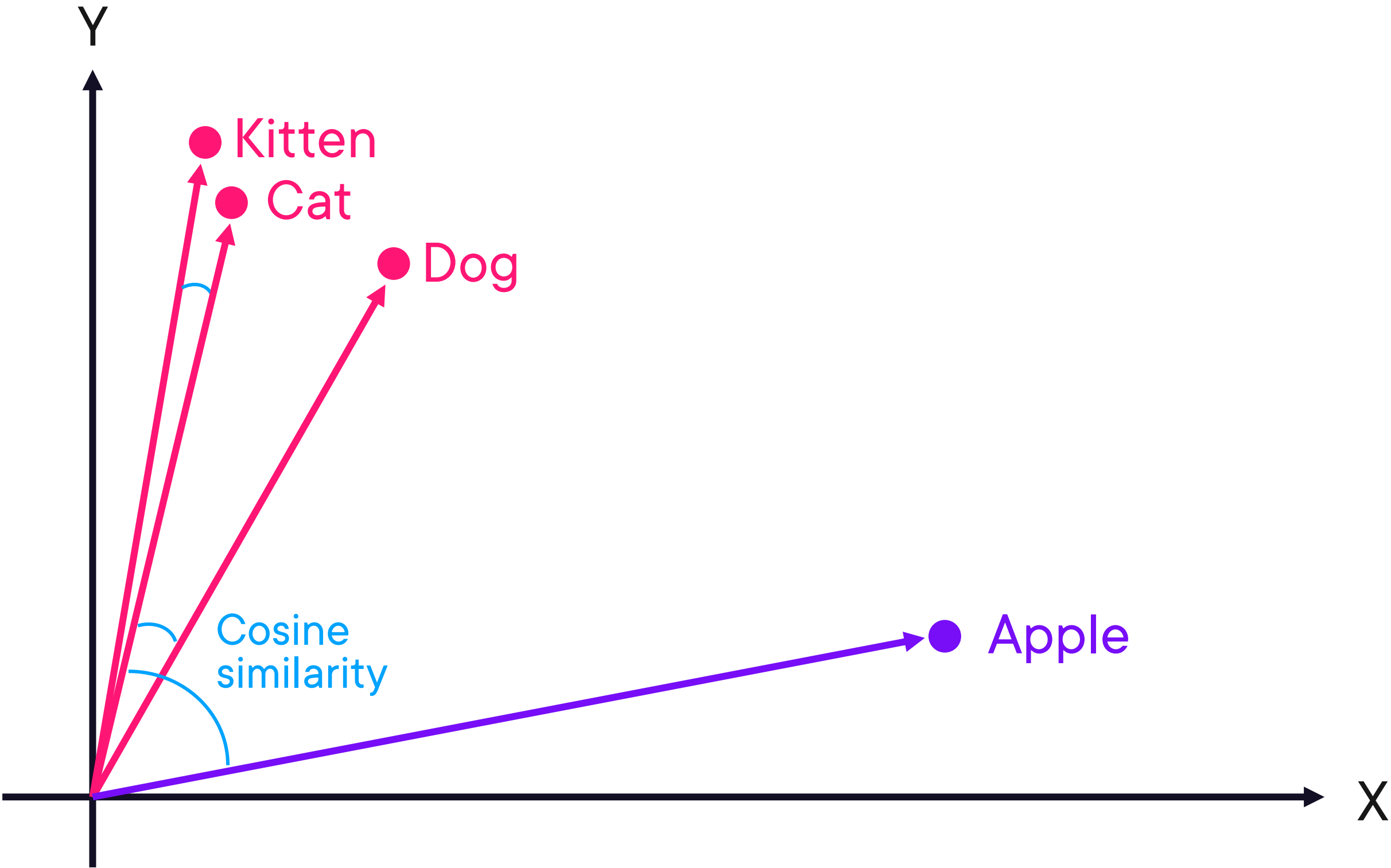
Representing Vectors



Vector Similarities



Vector Similarities

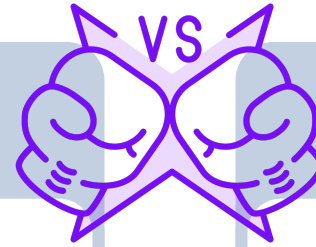




Understanding Vector Databases



Traditional vs. Vector Databases



Traditional databases

Based on SQL or NoSQL for data retrieval.

Organize data into tables, documents, or key-value pairs.

Efficient for exact match queries and operations like JOIN, SORT, and AGGREGATE.

Use B-trees, hash maps, or similar data structures for indexing.

Vector databases

Utilize vectors to represent complex, high-dimensional data.

Support similarity search in high-dimensional spaces (e.g., cosine similarity).

Employ approximate nearest neighbor (ANN) algorithms for efficient searching.

Ideal for unstructured data such as images, audio, text, and more.



Real-world Applications

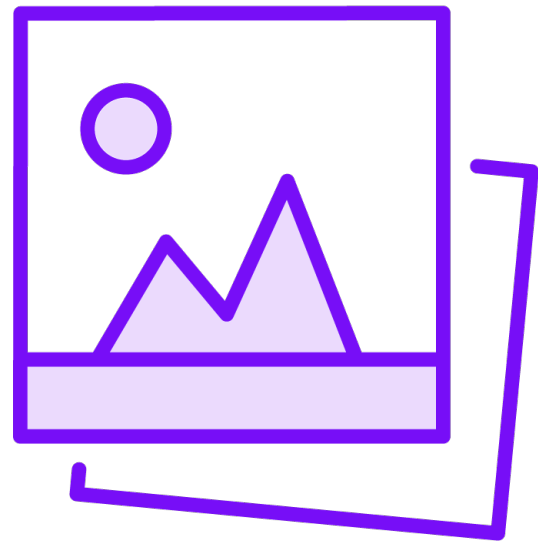
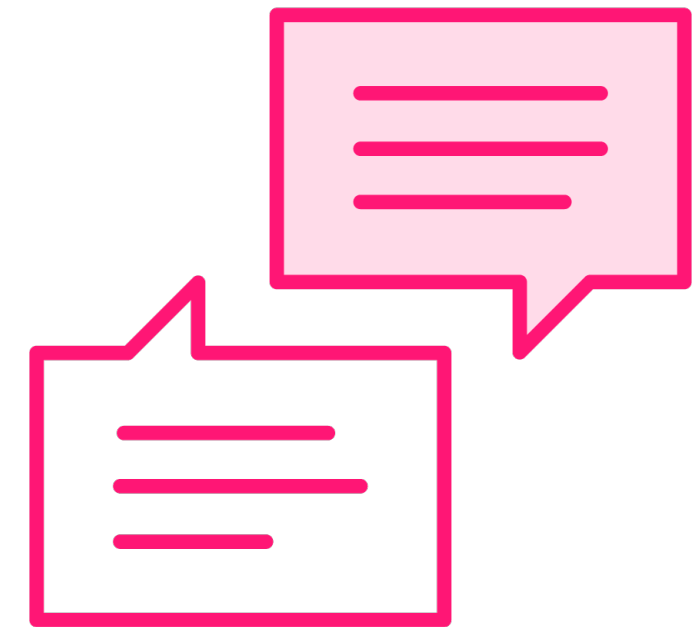


Image recognition



**Recommendation
engines**



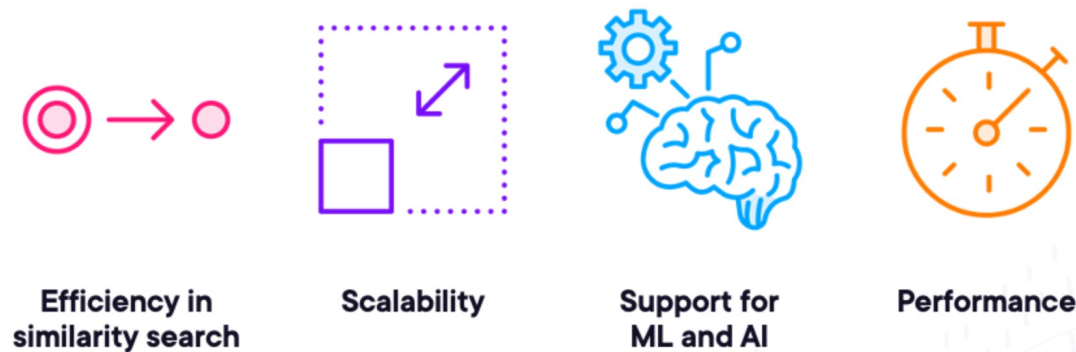
**Natural language
processing**

SWOT of Vector Databases



SWOT of Vector Databases

Strengths



Weaknesses



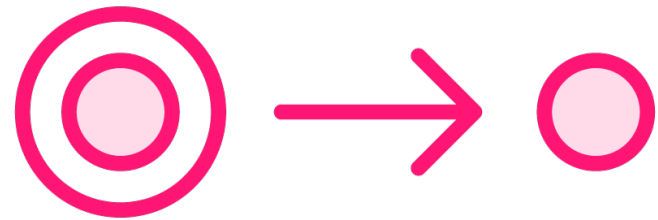
Opportunities



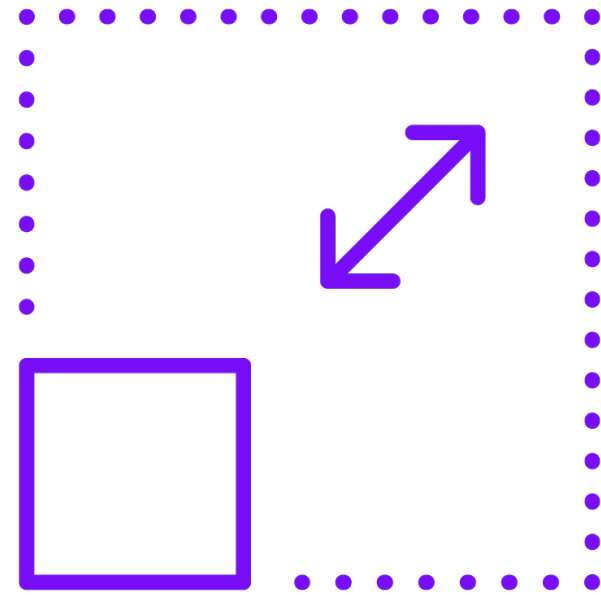
Threats



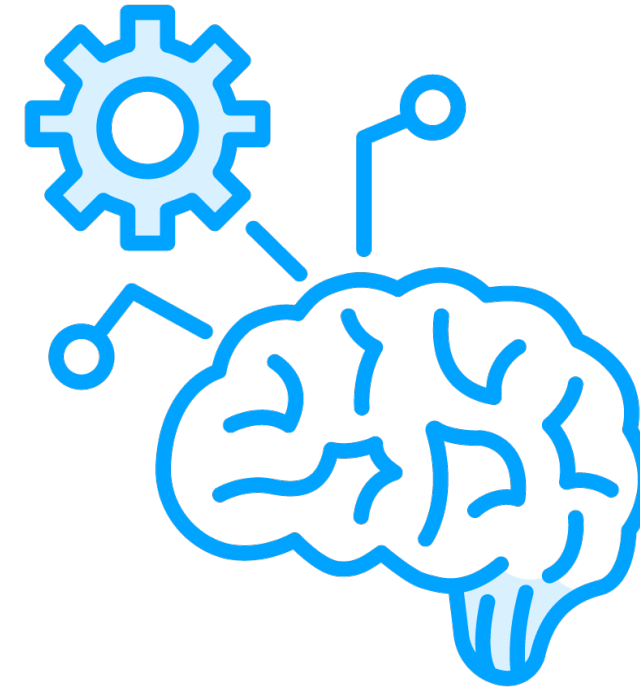
Strengths



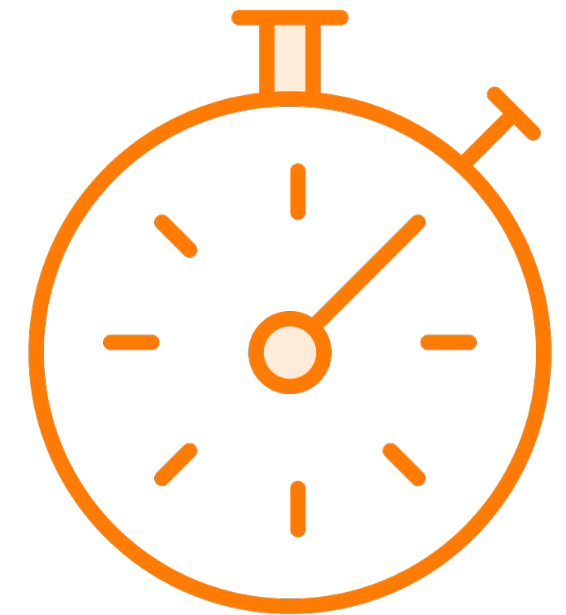
**Efficiency in
similarity search**



Scalability



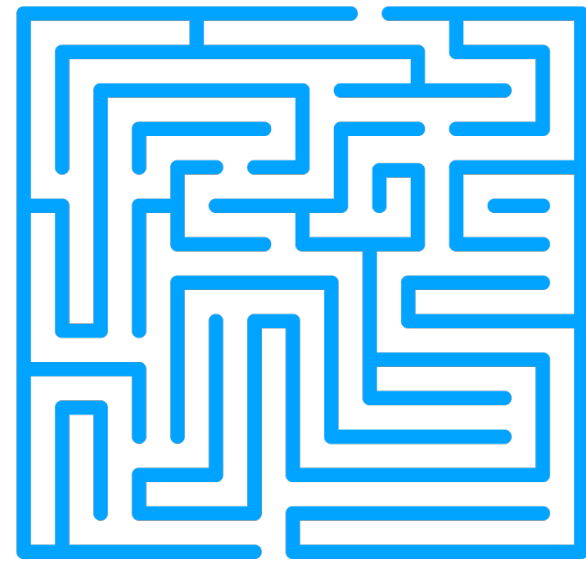
**Support for
ML and AI**



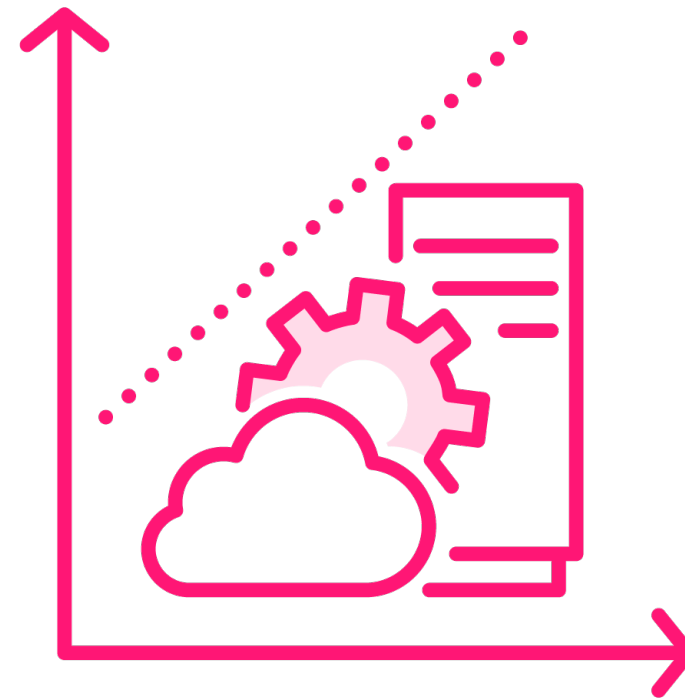
Performance



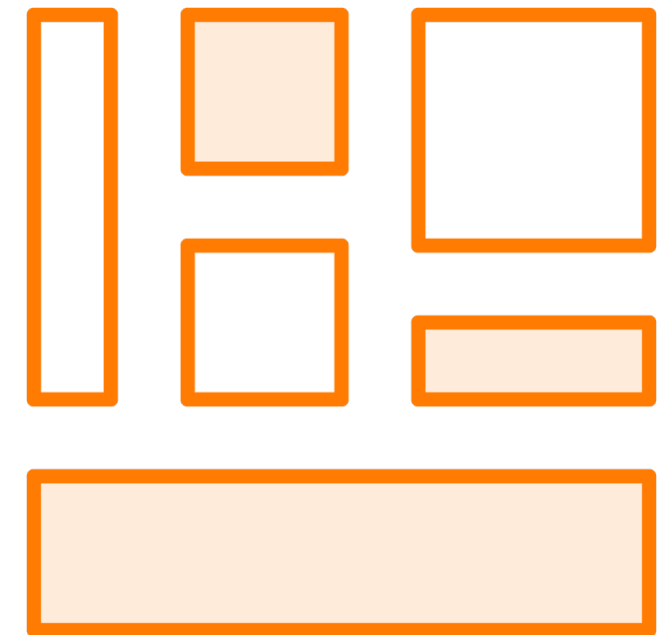
Weaknesses



Complexity

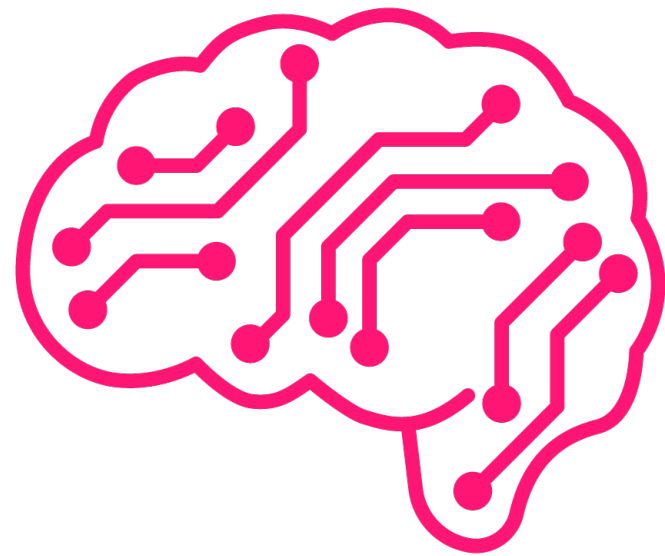


Resource intensive

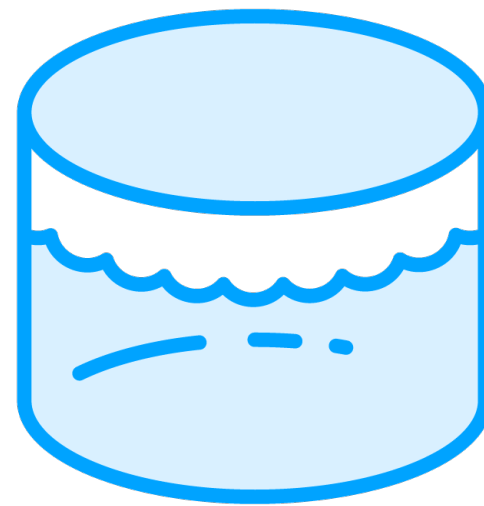


Limited query flexibility

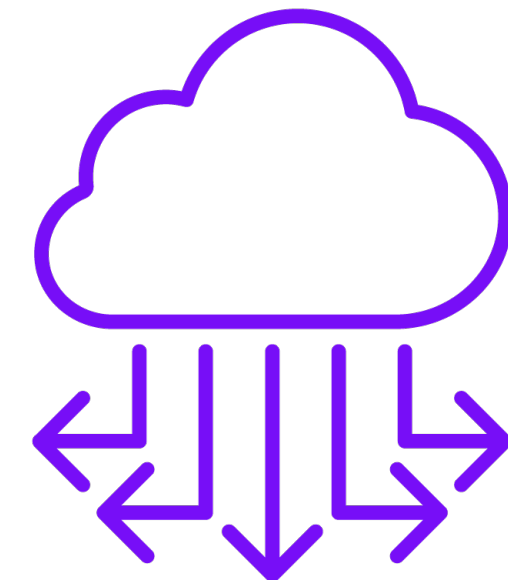
Opportunities



Growth of AI and ML



Big Data applications



**Integration with cloud
and edge computing**



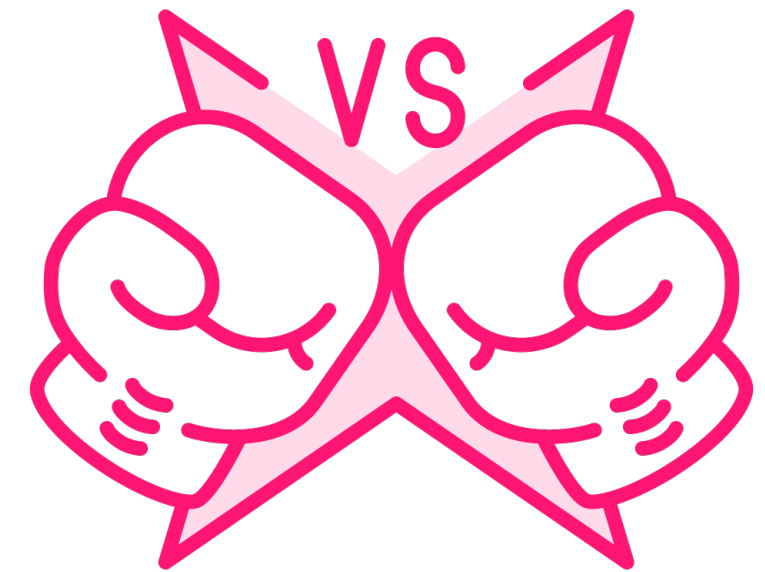
Threats



**Rapid technological
change**



**Security and privacy
concerns**



**Competition from
established database
solutions**

